

Stablecoins and the Exorbitant Privilege: A New Channel for Safe-Asset Demand and Its Systemic Fragility

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Yonsei GSIS — Topics in International Finance (2026-1)

April 22, 2026

1. The Idea and Its Motivation

This proposal asks whether large-scale USD-pegged stablecoins introduce a novel, two-sided form of systemic fragility — a “New Triffin Dilemma” — and, if so, whether the severity of the downside risk depends critically on the adequacy of stablecoin issuers’ reserve buffers relative to the velocity and scale of potential runs. The two questions are inseparable: understanding the fragility embedded in stablecoins’ structural role as Treasury buyers requires a theory of how large a reserve cushion is needed to prevent a programmatic Treasury liquidation from cascading into a sovereign-linked liquidity shock.

The motivation is both empirical and policy-driven. Total market capitalization of USD-pegged stablecoins surged from under \$50 billion in 2021 to over \$300 billion by early 2026, with Tether (USDT) and USD Coin (USDC) collectively backing their liabilities 1:1 with short-duration dollar assets, making their issuers among the largest mechanical buyers of short-term U.S. Treasuries. Projections envision a \$1–2 trillion ecosystem within this decade. This creates a structural two-sidedness: stablecoins deepen demand for U.S. safe assets in normal times (amplifying the “exorbitant privilege”), yet a run on a large issuer forces rapid Treasury liquidation, threatening a sovereign-linked liquidity shock. The U.S. Congress, the EU, Japan, and Hong Kong are each finalizing stablecoin reserve legislation, yet the central regulatory question — how large and how liquid must reserves be for a stablecoin ecosystem of a given scale to remain safe? — has no formal quantitative answer grounded in international finance theory. This proposal provides one.

2. State of the Literature

The theoretical foundation is Maggiori (2017), whose two-country continuous-time model characterizes the exorbitant privilege through an equilibrium risk-sharing condition in which the marginal value of rest-of-world (RoW) financial net worth — countercyclical and designated $\Omega^*(\tilde{N}^*)$ — is the key driver of safe-asset demand. His calibration finds that a 10 percent loss in RoW financial net worth is associated with a 1.2 percent deterioration in the U.S. net foreign asset position. Critically, his framework predates rule-bound programmatic Treasury buyers and contains no role for reserve adequacy in governing shock amplification — the gap this proposal addresses.

Gourinchas and Rey (2007) establish the empirical stylized facts of the privilege — the U.S. persistent trade deficit financed through valuation effects — and provide the measurement architecture this proposal extends to the stablecoin era. Caballero, Farhi, and Gourinchas (2008) show that the U.S.’s superior capacity to produce safe assets generates structural global imbalances; stablecoin issuers represent a new, non-sovereign safe-asset conduit that their framework did not anticipate.

Cole and Kehoe (2000) provide the most relevant model of the downside scenario: a self-fulfilling debt crisis that materializes when fundamentals enter a “crisis zone” defined by debt level and maturity. Their framework maps directly onto the stablecoin run: forced Treasury liquidation may

push yields high enough to validate the original redemption pressure. Crucially, their model implies a threshold structure — whether an issuer’s fundamentals lie inside or outside the crisis zone depends on buffer size.

On reserve adequacy, Jeanne and Rancière (2011) and Obstfeld, Shambaugh, and Taylor (2010) model optimal foreign exchange reserves as precautionary buffers against sudden stops. Their key insight — that the adequate reserve level is increasing in liability dollarization, financial openness, and the velocity of potential outflows — translates naturally to the stablecoin context, where liabilities are par-valued dollar claims subject to instantaneous, algorithmically-mediated redemption. We are the first to apply a reserve adequacy framework to stablecoin issuers as de facto quasi-sovereign entities. Gorton and Zhang (2023) compare stablecoins to wildcat banking, arguing runs are inherent without credible deposit insurance equivalents, but do not connect reserve adequacy to the international macro literature on safe-asset demand or model feedback from stablecoin stress to sovereign debt markets — the gap this proposal addresses.

3. Contribution

This proposal makes two connected contributions. Theoretically, we extend Maggiori’s (2017) open-economy framework by introducing stablecoin supply S_t and an explicit reserve buffer $B_t \equiv R_t - S_t$ — the excess of issuers’ liquid Treasury holdings over par liabilities. The modified equilibrium demand for U.S. safe assets becomes:

$$B^*(\tilde{N}^*, S, B) = B^*_{precautionary}(\tilde{N}^*) + B^*_{abnormal}(S) + B^*_{buffer}(B)$$

The buffer B_t enters asymmetrically. In normal times it over-demands Treasuries, amplifying the privilege. During a run (negative ΔS_t), a sufficiently large buffer absorbs redemptions without forced liquidation. When the buffer is inadequate — formally, when $|\Delta S_t| > B_t$ — the liquidation channel is activated and the Cole-Kehoe crisis-zone mechanism applies. This generates a testable “reserve adequacy threshold” below which a run of given speed and scale produces measurable sovereign spillovers, and above which it does not. No existing paper in international macro formalizes this asymmetric amplification-versus-absorption structure.

Empirically, we provide the first time-series evidence on whether stablecoin issuance growth Granger-causes compression in U.S. borrowing spreads, and whether sovereign-linked liquidity shocks during stablecoin stress events are moderated by the prevailing reserve buffer level — a buffer-conditioned event study novel in both the stablecoin and reserve adequacy literatures.

4. Methodology

4.1 Theoretical Extension

Empirical strategy. The dependent variable is the OIS-Treasury spread (3-month T-bill yield minus 3-month OIS rate, sourced from FRED), proxying the market premium for holding Treasuries over risk-free swaps — a direct demand-side measure of the privilege. We supplement with the Treasury bill auction bid-cover ratio (TreasuryDirect) as an alternative. The main regression, extending Maggiori’s Equation 21, is:

$$Spread_t = \alpha + \beta_1 \Delta \ln S_t + \beta_2 B_t + \beta_3 (B_t \times \Delta \ln S_t) + \beta_4 V_t + \beta_5 VIX_t + \beta_6 \Delta \ln N^*_t + \varepsilon_t$$

where S_t is total USD-pegged stablecoin market capitalization (DeFiLlama, monthly), B_t is the reserve buffer constructed as disclosed Treasury holdings minus outstanding supply scaled by supply (from Tether quarterly and Circle monthly attestations), and V_t is the 7-day rolling standard deviation of daily net supply changes — a direct measure of redemption velocity. The interaction term $B_t \times \Delta \ln S_t$ is the central innovation: under the privilege hypothesis, $\beta_1 < 0$ (issuance compresses spreads) and $\beta_3 > 0$ (a larger buffer dampens this effect). Standard errors are Newey-West (one lag).

Reserve adequacy threshold. We estimate the critical buffer level B^* below which run-induced Treasury liquidation generates statistically significant abnormal OIS-Treasury spread increases, using a Hansen (2000) threshold regression. This produces a data-driven analog to the Cole-Kehoe “crisis zone” directly applicable to regulatory minimum reserve requirements.

Event study. For the fragility hypothesis, we conduct a buffer-conditioned event study around three identified stress episodes: (i) the LUNA/UST collapse (May 9, 2022; low-buffer environment); (ii) the USDT partial depeg (May 12, 2022); and (iii) the USDC depeg following Silicon Valley Bank’s failure (March 11, 2023; higher-buffer environment for USDC). For each event, we compute mean abnormal changes in the OIS-Treasury spread over a $[-5, +20]$ trading-day window and test whether abnormal responses differ significantly across high- and low-buffer events.

Data. OIS spreads and T-bill yields: FRED API. Stablecoin supply and velocity: DeFiLlama stablecoins API (daily). Reserve composition: Tether quarterly attestations; Circle USDC monthly attestations. Auction bid-cover ratios: TreasuryDirect. VIX: CBOE via FRED. RoW financial net worth proxy: Datastream World ex-US equity index. Sample: January 2020 – March 2026 (~75 monthly observations). The short sample is an acknowledged constraint inherent to the post-2020 emergence of the stablecoin market and motivates the supplementary event study design.

5. References (added)

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